Multivariate Calculus Lecture Notes
Section 15.3 – Triple Integrals

I. Definition of a triple integral
   a. Suppose we want to find the mass of a three-dimensional solid W, where the continuous function $f(x, y, z)$ gives the density at the point (x, y, z) in W.
   b. We could slice W into little boxes, approximate the mass on each box, and then add up the masses.
   c. Graphically:

   d. The mass of the ijk^{th} box is approximately $f(P_{ijk})\Delta V_{ijk}$, where P_{ijk} is some point in the ijk^{th} box.
   e. Adding up these masses gives an approximation of the total mass of W:
$$\sum_i \sum_j \sum_k f(P_{ijk})\Delta V_{ijk}$$
   f. And if we let the volume of the largest box go to zero we get the exact value of the mass of W:
$$\lim_{||P||\to 0} \sum_i \sum_j \sum_k f(P_{ijk})\Delta V_{ijk}.$$
   g. This limit is defined to be the triple integral of f over W:
$$\iiint_W f(x, y, z)\, dV = \lim_{||P||\to 0} \sum_i \sum_j \sum_k f(P_{ijk})\Delta V_{ijk}.$$

II. Comments
   a. Volume differential in rectangular coordinates
      i. We have $dV = dx\, dy\, dz$.
      ii. This is because a volume differential in rectangular coordinates is a box, and the volume of a box is length $\times$ width $\times$ height.
   b. Some interpretations of the triple integral
      i. The standard interpretation of $\iiint_W f(x, y, z)\, dV$ is the mass of W, where f is the density function of the solid W (and thus must be positive over W).
      ii. The expression $\iiint_W dV$ gives the volume of W, so that you can find volumes with triple integrals just as you can with double integrals.
      iii. You can also use $\iiint_W f(x, y, z)\, dV$ to find average value, center of mass, and moments of inertia. (See Section 15.5.)

III. Examples
   a. Use a triple integral to find the volume of the tetrahedron bounded by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$, $z = 0$.

i. Graphically:

ii. The volume will be $\iiint_W dV$, where W is the tetrahedron.

iii. The region is "z-simple": Every straight up-and-down line starts at the same function and ends at the same function.
   1. Thus it is probably easiest to integrate with respect to z first.
   2. Lower bound: $z = 0$
   3. Upper bound: $z = 2 - x - 2y$

iv. Now that we've taken care of z, we have to figure out the bounds on the x and y variables.
   1. Project the graph onto the other two dimensions:

   2. This region is vertically but not horizontally simple. So we should do a y integration next.
      - Lower bound: $y = x/2$
      - Upper bound: $y = 1 - x/2$
   3. Then the x bounds are $x = 0$ and $x = 1$.

v. Thus our integral is $\displaystyle\int_0^1 \int_{x/2}^{1-x/2} \int_0^{2-x-2y} dz\, dy\, dx$.

vi. We have

$$\begin{aligned}
\int_0^1 \int_{x/2}^{1-x/2} \int_0^{2-x-2y} dz\, dy\, dx &= \int_0^1 \int_{x/2}^{1-x/2} z\Big|_0^{2-x-2y} dy\, dx = \int_0^1 \int_{x/2}^{1-x/2} (2-x-2y)\, dy\, dx \\
&= \int_0^1 2y - xy - y^2\Big|_{x/2}^{1-x/2} dx \\
&= \int_0^1 \left(2 - x - x + \frac{x^2}{2} - \left(1 - \frac{x}{2}\right)^2 - x + \frac{x^2}{2} + \frac{x^2}{4}\right) dx \\
&= \int_0^1 (1 - 2x + x^2)\, dx = x - x^2 + \frac{1}{3}x^3\Big|_0^1 = 1 - 1 + \frac{1}{3} = \frac{1}{3}.
\end{aligned}$$

b. Evaluate the integral $\iiint_Q z\, dV$, where Q is the solid bounded by $y^2 + z^2 = 9$ and the planes $x = 0$, $y = 3x$, $z = 0$, in the first octant. (If z is a density function, we could think of this integral as yielding the mass of Q.)

i. First, graph the solid:

ii. The region is x-simple, y-simple, and z-simple.

iii. Let's set up the problem three different ways.

iv. Integrate with respect to z first

1. Bounds on z: $z = 0$ to $z = \sqrt{9 - y^2}$
2. Then project solid onto the xy-plane:

3. If we do y next we have y from $3x$ to 3.
4. Then the x bounds are from 0 to 1.
5. This gives us the triple integral $\displaystyle\int_0^1 \int_{3x}^3 \int_0^{\sqrt{9-y^2}} z \, dz \, dy \, dx$.

v. Integrate with respect to y first

1. Bounds on y: $y = 3x$ to $y = \sqrt{9 - z^2}$
2. Then project solid onto xz-plane.
   - What does this look like?
   - The boundary is the intersection of the plane $y = 3x$ and the cylinder $y^2 + z^2 = 9$.
   - The relationship between x and z is thus $9x^2 + z^2 = 9$.
   - And that's all we're interested in with the projection, as the projection onto the xz-plane ignores what's going on with y.
   - The equation $9x^2 + z^2 = 9$ is that of an ellipse.
   - Since x and z must both be positive, the projection looks like

3. If we integrate next with respect to z the bounds are from 0 to $\sqrt{9 - 9x^2}$.
4. Then the x bounds are from 0 to 1.
5. Thus the integral is $\displaystyle\int_0^1 \int_0^{\sqrt{9-9x^2}} \int_{3x}^{\sqrt{9-z^2}} z \, dy \, dz \, dx$.

vi. Integrate with respect to x first

1. Bounds on x: $x = 0$ to $x = y/3$
2. Then project solid onto yz-plane:

3. If we integrate next with respect to z the bounds are from 0 to $\sqrt{9 - y^2}$.
4. Then the y bounds are from 0 to 3.
5. Thus the integral is $\displaystyle\int_0^3 \int_0^{\sqrt{9-y^2}} \int_0^{y/3} z \, dx \, dz \, dy$.

vii. The last one is probably easiest to evaluate, so let's try that one.

$$\int_0^3 \int_0^{\sqrt{9-y^2}} \int_0^{y/3} z \, dx \, dz \, dy = \int_0^3 \int_0^{\sqrt{9-y^2}} xz \Big|_{x=0}^{y/3} dz \, dy = \frac{1}{3} \int_0^3 \int_0^{\sqrt{9-y^2}} yz \, dz \, dy$$

$$= \frac{1}{6}\int_0^3 yz^2\Big|_{z=0}^{\sqrt{9-y^2}}\,dy = \frac{1}{6}\int_0^3 (9y - y^3)\,dy = \frac{1}{6}\left[\frac{9}{2}y^2 - \frac{1}{4}y^4\right]_0^3$$
$$= \frac{1}{6}\left(\frac{81}{2} - \frac{81}{4}\right) = \frac{27}{8}.$$

c. Draw the region of integration given by the triple integral $\displaystyle\int_0^1\int_y^1\int_0^{\sqrt{1-y^2}} dz\,dx\,dy$.

i. Let's work backwards.
ii. Lastly we integrate y from 0 to 1:
iii. Second, we integrate x from y to 1:

iv. First, we integrate z from 0 to $\sqrt{1-y^2}$:

Homework: Section 15.3 #6, 13, 18, 19, 20, 24, 29, 30.